

Introduction to Linear Optimization and Extensions with MATLAB

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Chapter 4

Duality

Motivation for Duality

- Consider following LP:

$$\begin{array}{ll}\text{maximize} & 6x_1 + x_2 + 2x_3 \\ \text{subject to} & x_1 + 2x_2 - x_3 \leq 20 \\ & 2x_1 - x_2 + 2x_3 \leq 30 \\ & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0\end{array}$$

- A lower bound of the LP can be found by finding any feasible solution, e.g. $x_1 = 10$, $x_2 = 2$, $x_3 = 2$ will result in $z = 66 \leq z^*$.
- How can we get an upper bound?

Motivation for Duality

$$\begin{aligned} &\text{maximize} && 6x_1 + x_2 + 2x_3 \\ &\text{subject to} && x_1 + 2x_2 - x_3 \leq 20 \\ & && 2x_1 - x_2 + 2x_3 \leq 30 \\ & && x_1 \geq 0, x_2 \geq 0, x_3 \geq 0 \end{aligned}$$

- If multiplying 2 for both constraints and add them together i.e. $2(x_1 + 2x_2 - x_3) + 2(2x_1 - x_2 + 2x_3) = 6x_1 + 2x_2 + 2x_3 \leq 2(20) + 2(30) = 100$, note that $6x_1 + x_2 + 2x_3 = z \leq 6x_1 + 2x_2 + 2x_3 \leq 100$ for any $x_1, x_2, x_3 \geq 0$, we have $z^* \leq 100$.
- $66 \leq z^* \leq 100$. Ideally, we want shrink the gap between upper bound and lower bound to 0.

Motivation for Duality

- Generally, we multiply non-negative variables π_1, π_2 to the constraints and add them together:

$$(\pi_1)(x_1 + 2x_2 - x_3) + (\pi_2)(2x_1 - x_2 + 2x_3) \leq (\pi_1)20 + (\pi_2)30$$

$$(\pi_1 + 2\pi_2)x_1 + (2\pi_1 - \pi_2)x_2 + (-\pi_1 + 2\pi_2)x_3 \leq 20\pi_1 + 30\pi_2$$

- To achieve a minimum upper bound we require that:
 - (1) $\pi_1 + 2\pi_2 \geq 6, 2\pi_1 - \pi_2 \geq 1, -\pi_1 + 2\pi_2 \geq 2$
 - (2) minimize $20\pi_1 + 30\pi_2$
 - (3) $\pi_1 \geq 0, \pi_2 \geq 0$
- (1)(2)(3) can be formulated as an LP which is called the dual of the primal problem.

Primal-Dual Pair

$$\begin{array}{ll} \text{maximize} & 6x_1 + x_2 + 2x_3 \\ \text{subject to} & x_1 + 2x_2 - x_3 \leq 20 \\ & 2x_1 - x_2 + 2x_3 \leq 30 \\ & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, \end{array}$$

$$\begin{array}{ll} \text{minimize} & 20\pi_1 + 30\pi_2 \\ \text{subject to} & \pi_1 + 2\pi_2 \geq 6 \\ & 2\pi_1 - \pi_2 \geq 1 \\ & -\pi_1 + 2\pi_2 \geq 2 \\ & \pi_1 \geq 0, \pi_2 \geq 0, \end{array}$$

$$\begin{array}{ll} \text{maximize} & \sum_{i=1}^n c_i x_i \\ \text{subject to} & \sum_{j=1}^n a_{ij} x_j \leq b_i \quad i = 1, \dots, m \\ & x_j \geq 0 \quad j = 1, \dots, n, \end{array}$$

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^m b_i \pi_i \\ \text{subject to} & \sum_{i=1}^m a_{ij} \pi_i \geq c_j \quad j = 1, \dots, n \\ & \pi_j \geq 0 \quad j = 1, \dots, m. \end{array}$$

$$\begin{array}{ll} \text{maximize} & c^T x \\ \text{subject to} & Ax \leq b \\ & x \geq 0. \end{array}$$

$$\begin{array}{ll} \text{minimize} & b^T \pi \\ \text{subject to} & A^T \pi \geq c \\ & \pi \geq 0. \end{array}$$

Primal problem (P)

Dual problem (D)

Forming the dual for general LP

- The following tables give the relationship between a primal-dual pair of linear programs:

(P)	$\Leftrightarrow$	(D)
Constraint matrix $\mathbf{A}$	$\Leftrightarrow$	Constraint matrix $\mathbf{A}^T$
Cost vector $\mathbf{c}$	$\Leftrightarrow$	Cost vector $\mathbf{b}$
RHS $\mathbf{b}$	$\Leftrightarrow$	RHS $\mathbf{c}$

maximization	$\Leftrightarrow$	minimization
variables	$\Leftrightarrow$	constraints
≥ 0	$\Leftrightarrow$	$\geq$
≤ 0	$\Leftrightarrow$	$\leq$
unrestricted	$\Leftrightarrow$	$=$
constraints	$\Leftrightarrow$	variables
$\geq$	$\Leftrightarrow$	≤ 0
$\leq$	$\Leftrightarrow$	≥ 0
$=$	$\Leftrightarrow$	unrestricted

Example: Dual Formulation

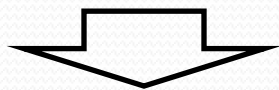
• (P)

maximize $5x_1 - 3x_2 + 2x_3$
subject to $x_1 + 4x_2 - 2x_3 \geq 1$

$$2x_1 - x_2 + x_3 = -5$$

$$x_1 + x_2 - x_3 \leq 2$$

$x_1 \leq 0, x_2 \geq 0, x_3$ unrestricted



$$c = \begin{bmatrix} 5 \\ -3 \\ 2 \end{bmatrix}, b = \begin{bmatrix} 1 \\ -5 \\ 2 \end{bmatrix}, A = \begin{bmatrix} 1 & 4 & -2 \\ 2 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

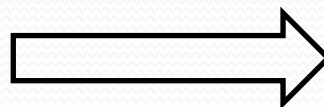
(D)

minimize $\pi_1 - 5\pi_2 + 2\pi_3$
subject to $\pi_1 + 2\pi_2 + \pi_3 \leq 5$

$$4\pi_1 - \pi_2 + \pi_3 \geq -3$$

$$-2\pi_1 + \pi_2 - \pi_3 = 2$$

$\pi_1 \leq 0, \pi_2$ unrestricted, $\pi_3 \geq 0$



$$A^T = \begin{bmatrix} 1 & 2 & 1 \\ 4 & -1 & 1 \\ -2 & 1 & -1 \end{bmatrix}$$

Duality Theory

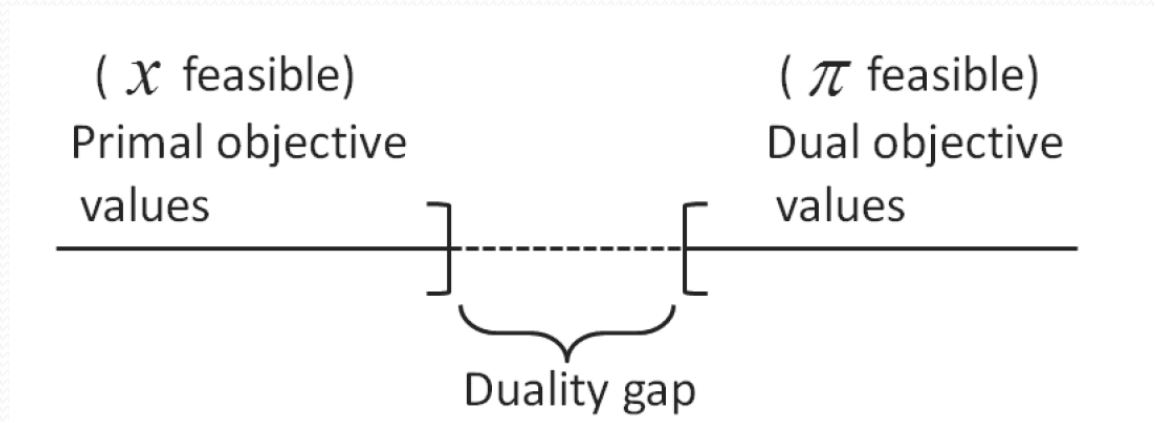
- Theorem 4.2 (Weak Duality):

Let $x = (x_1, \dots, x_n)^T$ be a feasible solution for a primal problem P and $\pi = (\pi_1, \dots, \pi_m)^T$ be a feasible solution for a dual problem D. Then, $\sum_{j=1}^n c_j x_j \leq \sum_{i=1}^m b_i \pi_i$.

- Proof:

$$\sum_{j=1}^n c_j x_j \leq \sum_{j=1}^n \left(\sum_{i=1}^m a_{ij} \pi_i \right) x_j = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} x_j \right) \pi_i \leq \sum_{i=1}^m b_i \pi_i$$

Duality Gap

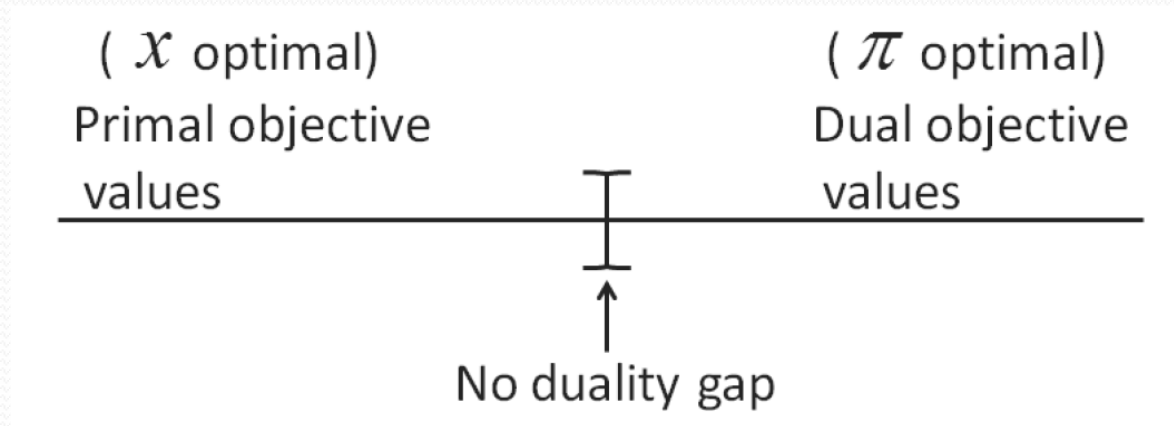


Duality Theory

Corollary 4.1: Let $x^* = (x_1^*, \dots, x_n^*)^T$ be a feasible solution for a primal problem P and $\pi^* = (\pi_1^*, \dots, \pi_m^*)^T$ be a feasible solution for a dual problem D, and $c^T x^* = b^T \pi^*$, then x^* and π^* are optimal solutions for their respective problems.

- Corollary 4.2: If the primal problem P is unbounded, then the dual problem D is infeasible.
- Corollary 4.3: If the dual problem D is unbounded, then the primal problem P is infeasible.

No Duality Gap



Duality Theory

- Theorem 4.3 (Strong Duality):

If a linear programming problem has a finite optimal solution, then its dual problem will also have a finite optimal solution and the optimal objective function values of both problems are the same.

- Proof: Let P is in standard form and corresponding D is

minimize $c^T x$

subject to $Ax = b$ $\Leftrightarrow$

$x \geq 0$

maximize $b^T \pi$

subject to $A^T \pi \leq c$

π unrestricted

P has been solved by Simplex Method to generate an optimal basic feasible solution $x^* = [x_B^* \ x_N^*]^T$, and the corresponding reduced cost is non-negative i.e. $r_N = c_N - c_B^T B^{-1} N \geq 0$.

Duality Theory

- Now let $\pi^* = (c_B^T B^{-1})^T$, then

$$c - A^T \pi^* = \begin{bmatrix} c_B \\ c_N \end{bmatrix} - \begin{bmatrix} B^T \\ N^T \end{bmatrix} \pi^* = \begin{bmatrix} c_B \\ c_N \end{bmatrix} - \begin{bmatrix} B^T \\ N^T \end{bmatrix} (c_B^T B^{-1})^T = \begin{bmatrix} c_B \\ c_N \end{bmatrix} - \begin{bmatrix} c_B \\ (B^{-1}N)^T c_B \end{bmatrix} = \begin{bmatrix} 0 \\ r_N \end{bmatrix} \geq 0$$

thus π^* is a feasible solution for D. Furthermore,

$$b^T \pi^* = (\pi^*)^T b = c_B^T B^{-1} b = c_B^T x_B^* = c^T x^*$$

by Corollary 4.1, π^* is an optimal solution for the dual.

Primal-Dual Possibilities

- Combined with weak duality, corollaries 4.1 – 4.3, and strong duality, following primal-dual pair can be stated:

	optimal	infeasible	unbounded
optimal	yes	no	no
infeasible	no	yes	yes
unbounded	no	yes	no

Complementary Slackness

- Consider canonical primal-dual pair of LP:

$$\begin{array}{ll} \text{(P) maximize } c^T x & \text{(D) minimize } b^T \pi \\ \text{subject to } Ax \leq b & \text{subject to } A^T \pi \geq c \\ x \geq 0 & \pi \geq 0 \end{array}$$

- Let $x_s = b - Ax \geq 0$ be the primal slack vector,
 $\pi_s = A^T \pi - c \geq 0$ be the dual slack vector.

For any primal feasible x and dual feasible π , $x^T \pi_s + x_s^T \pi = (\pi^T A - c^T)x + \pi^T(b - Ax) = \pi^T b - c^T x = 0$ (duality gap), which means:

$$x^T \pi_s = 0 \text{ and } x_s^T \pi = 0$$

Complementary Slackness

- Theorem 4.4 (Complementary Slackness):

Let x be a feasible solution for the canonical primal problem P and π be a feasible solution for its dual D . Then, x and π are optimal solutions if and only if:

(1) $x_i(A^T\pi - c)_i = 0$, for $i = 1, \dots, n$

(2) $(b - Ax)_j\pi_j = 0$, for $j = 1, \dots, m$

Example: Complementary Slackness

- (P) $\max 50x_1 + 45x_2 + 30x_3$ (D) $\min 45\pi_1 + 25\pi_2 + 9\pi_3$
s.t. $4x_1 + 3x_2 + x_3 \leq 45$ s.t. $4\pi_1 + 2\pi_2 + 2\pi_3 \geq 50$
 $2x_1 + 2x_2 + 2x_3 \leq 25$ $3\pi_1 + 2\pi_2 + \pi_3 \geq 45$
 $2x_1 + x_2 + .5x_3 \leq 9$ $\pi_1 + 2\pi_2 + .5\pi_3 \geq 30$
 $x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$ $\pi_1 \geq 0, \pi_2 \geq 0, \pi_3 \geq 0$
- For (P), $x^* = (x_1^*, x_2^*, x_3^*)^T = (0, 5.5, 7)^T$, and $z_P^* = 457.5$.
Let slack variable vector $x_s^T = (x_4, x_5, x_6) = (21.5, 0, 0)^T$, and associated dual variable is $\pi_s = (\pi_1, \pi_2, \pi_3) = (0, 7.5, 30)^T$.
- For (D), $\pi^* = (\pi_1^*, \pi_2^*, \pi_3^*)^T = (0, 7.5, 30)^T$, and $z_D^* = 457.5$.
Let slack variable vector $\pi_s^T = (\pi_4, \pi_5, \pi_6) = (25, 0, 0)^T$, and associated dual variable is $x_s = (x_1, x_2, x_3) = (0, 5.5, 7)^T$.

Complementary Slackness for Standard Form

- Consider primal in standard form and its dual:

(P) maximize $c^T x$

subject to $Ax = b$

$x \geq 0$

(D) minimize $b^T \pi$

subject to $A^T \pi \leq c$

π unrestricted

- Theorem 4.5:

Suppose a primal problem P is in standard form. Then a vector x^* is optimal for P if and only if:

(1) $Ax^* = b$ and $x^* \geq 0$. (primal feasibility)

(2) There are vectors π^* and π_s such that $A^T \pi^* + \pi_s = c$
and $\pi_s \geq 0$. (dual feasibility)

(3) $x^T \pi_s = 0$. (complementary slackness)

Complementary Slackness for Standard Form

- Corollary 4.4:

The vectors π^* is an optimal solution for (D).

E.g. (P) $\min x_1 + x_2 + x_3 + x_4$ s.t. $x_1 + x_2 + x_3 = 4$ $2x_1 + x_2 + x_4 = 6$ $x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0$	(D) $\max 4\pi_1 + 6\pi_2$ s.t. $\pi_1 + 2\pi_2 \leq 1$ $\pi_1 + \pi_2 \leq 1$ $\pi_1 \leq 3$ $\pi_2 \leq 1$ π_1, π_2 unrestricted
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- For solution $x = (x_1, x_2, x_3, x_4)^T = (2, 2, 0, 0)^T$:
since $x_1, x_2 \neq 0$, $\pi_1 + 2\pi_2 = 1$, $\pi_1 + \pi_2 = 1 \Rightarrow \pi = (\pi_1, \pi_2)^T = (1, 0) \Rightarrow z_D = 4 = z_P \Rightarrow x$ is optimal for (P) and π is optimal for (D)

Economic Interpretation for the Dual

- E.g. produce table and chair with following parameter:

resources (units)	table	chair	total resources
wood (sq meters)	6	4	1200
cutting (hrs)	5	2	1000
finishing (hrs)	4	1	500
market price	\$12	\$7	

- $$(P) \max 12x_1 + 7x_2$$

$$\text{s.t. } 6x_1 + 4x_2 \leq 1200$$

$$8x_1 + 2x_2 \leq 1000$$

$$2x_1 + x_2 \leq 500$$

$$x_1 \geq 0, x_2 \geq 0$$

$$\Rightarrow x_1^* = 80, x_2^* = 180.$$

- $$(D) \min 1200\pi_1 + 1000\pi_2 + 500\pi_3$$

$$\text{s.t. } 6\pi_1 + 8\pi_2 + 2\pi_3 \geq 12$$

$$4\pi_1 + 2\pi_2 + \pi_3 \geq 7$$

$$\pi_1 \geq 0, \pi_2 \geq 0, \pi_3 \geq 0$$

$$\Rightarrow \pi_1^* = 1.6, \pi_2^* = 0.3, \pi_3^* = 0$$

Economic Interpretation for the Dual

- Consider that the manufacturer obtains the resources from a supplier, and he wish know the unit price, π_i , of each resource, and so minimize the total cost of obtaining these resources i.e. the objective of dual problem.
- Assuming the parameter information is public, then the dual constraints ensure that the total cost to produce a product (e.g. table) is at least the revenue obtained from the product (e.g. table).
- If total cost to unit product is grater than market price, according to the complementary slackness condition, such product will not be produced.

Economic Interpretation for the Dual

- The optimal dual prices can be interpreted as marginal prices or shadow prices indicating the value of extra unit of each resource.
- For example, observe that the first two primal constraints are tight at the primal optimality, which means that both the wood and cutting resource are exhausted at the optimal production plan. The optimal dual variables associated with these constraints are $\pi_1^* = 1.6$, $\pi_2^* = 0.3$, thus an additional unit of wood (cutting) will increase the revenue by \$1.6 (\$0.3). $\pi_3^* = 0$ means an extra unit of finishing does not increase the revenue.